

General-Purpose Input/Output (GPIO) Module

This chapter describes the general-purpose input/output (GPIO) module. The GPIO module provides the family of devices with input/output (I/O) capability. The I/O pins are bidirectional and bit-programmable. The GPIO module also supports external interrupt capability.

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25.1 Overview

The GPIO module offers general-purpose input and output capability. It supports up to eight 8-bit ports for a total of up to 64 GPIO terminals. Each of these 64 terminals can be independently configured as input or output and configured as required by the application. The GPIO module also supports generation of interrupts whenever a rising edge or falling edge or any toggle is detected on up to 32 of these GPIO terminals. Refer to the device datasheet for identifying the number of GPIO ports supported and the GPIO terminals capable of generating an interrupt.

The main features of the GPIO module are summarized as follows:

- Allows each GPIO terminal to be configured for general-purpose input or output functions
- Supports programmable pull directions on each input GPIO terminal
- Supports GPIO output in push/pull or open-drain modes
- Allows up to 32 GPIO terminals to be used for generating interrupt requests

25.2 Quick Start Guide

The GIO module comprises two separate components: an input/output (I/O) block and an interrupt generation block. [Figure 25-1](#) and [Figure 25-2](#) show what you should do after reset to configure the GIO module as I/O or for generating interrupts.

In GIO interrupt service routine, you shall read the GIO offset register (GIOFF1 or GIOFF2, depending on high-/low-level interrupt) to clear the flag and find the pending interrupt GIO channel.

Figure 25-1. I/O Function Quick Start Flow Chart

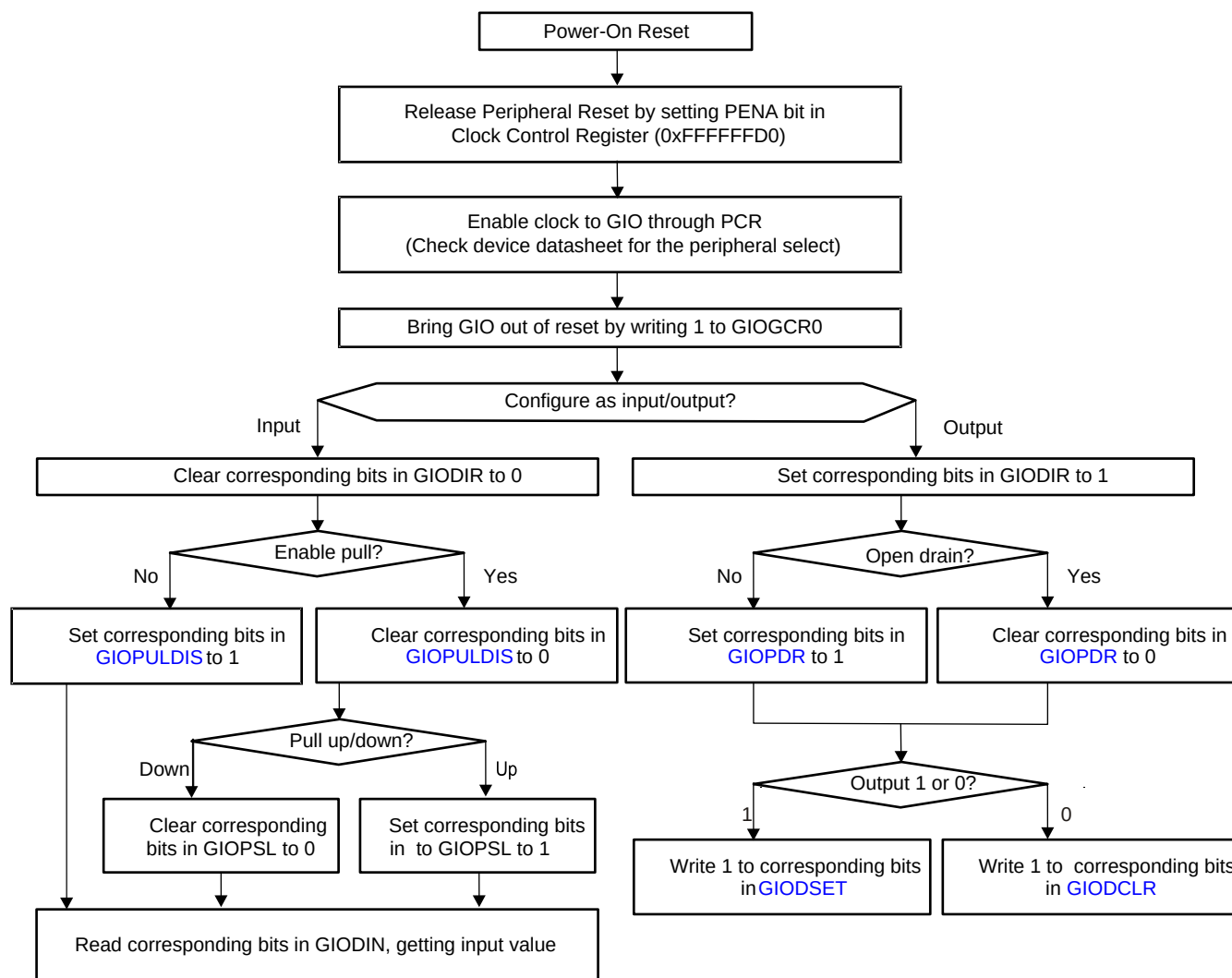
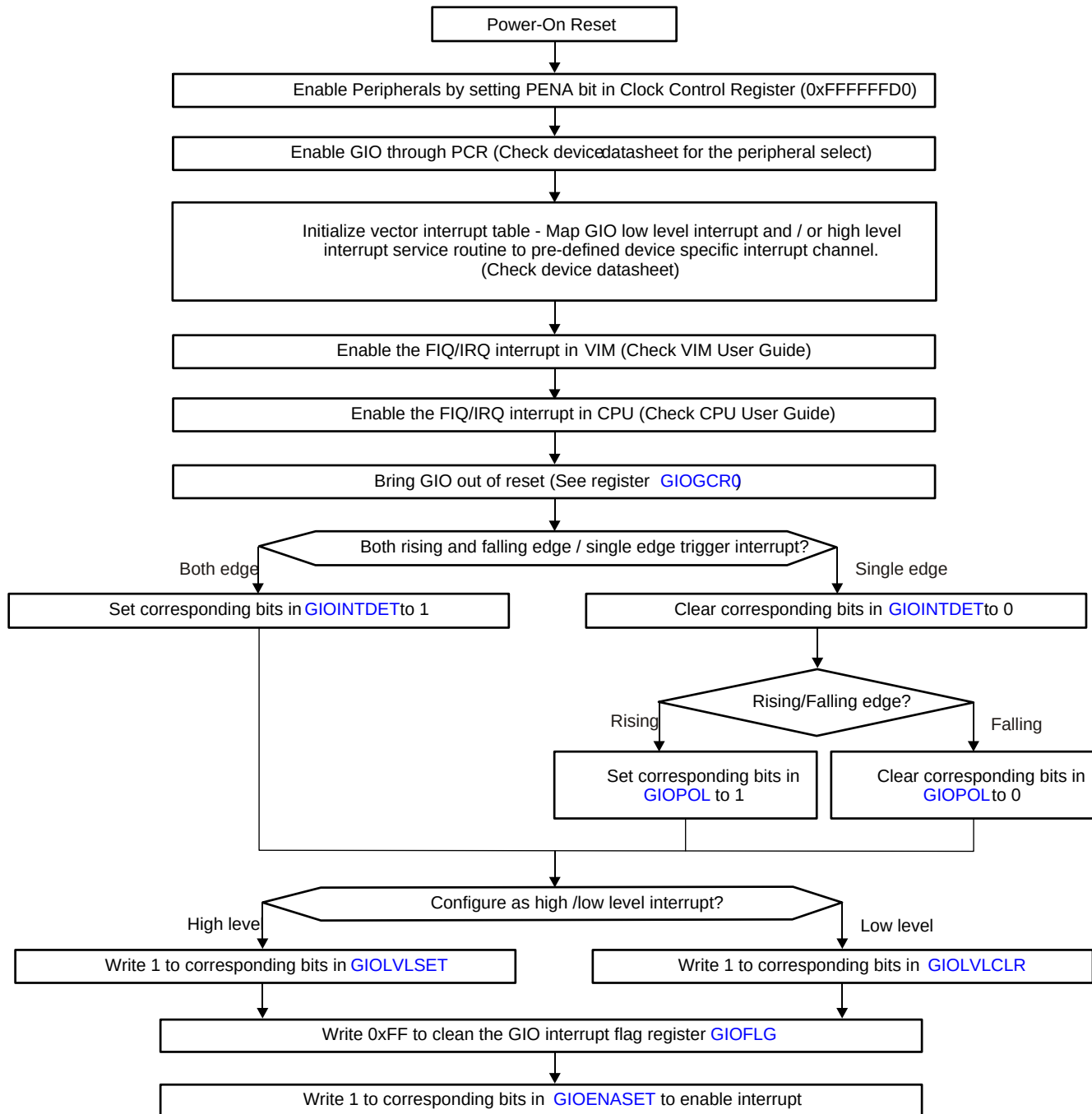
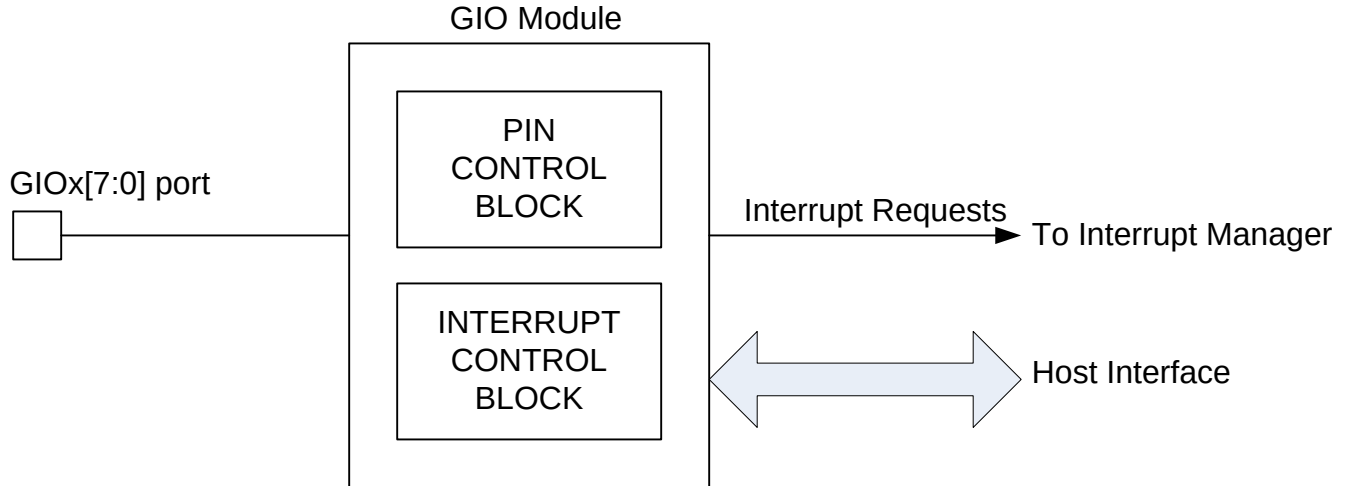


Figure 25-2. Interrupt Generation Function Quick Start Flow Chart


25.3 Functional Description of GIO Module

As shown in Figure 25-3, the GIO module comprises of two separate components: an input/output (I/O) block and an interrupt block.

Figure 25-3. GIO Module Diagram



25.3.1 I/O Functions

The I/O block allows each GIO terminal to be configured for use as a general-purpose input or output in the application. The GIO module supports multiple registers to control the various aspects of the input and output functions. These are described as follows.

- **Data direction (GIDIR)**
Configures GIO terminal(s) as input (default) or output through the GIDIRx registers.
- **Data input (GIDIN)**
Reflects the logic level on GIO terminals in the GIDINx registers. A high voltage (V_{IH} or greater) applied to the pin causes a high value (1) in the data input register (GIDIN[7:0]). When a low voltage (V_{IL} or less) is applied to the pin, the data input register reads a low value (0). The V_{IH} and V_{IL} values are device specific and can be found in the device datasheet.
- **Data output (GIDOUT)**
Configures the logic level to be output on GIO terminal(s) configured as outputs. A low value (0) written to the data output register forces the pin to a low output voltage (V_{OL} or lower). A high value (1) written to the data output register (GIDOUTx) forces the pin to a high output voltage (V_{OH} or higher) if the open drain functionality is disabled (GIOPDRx[7:0]). If open drain functionality is enabled, a high value (1) written to the data output register forces the pin to a high-impedance state (Z).
- **Data set (GIDSET)**
Allows logic HIGH to be output on GIO terminal(s) configured as outputs by writing 1's to the required bits in the GIDSETx registers. If open drain functionality is enabled, a high value (1) written to the data output register forces the pin to a high-impedance state (Z). The GIDSETx registers eliminate the need for the application to perform a read-modify-write operation when it needs to set one or more GIO pin(s).
- **Data clear (GIDCLR)**
Allows logic LOW to be output on GIO terminal(s) configured as outputs by writing 1s to the required bits in the GIDCLR registers. The GIDCLR registers eliminate the need for the application to perform a read-modify-write operation when it needs to clear one or more GIO pin(s).
- **Open drain (GIOPDR)**
Open drain functionality is enabled or disabled (default) using the open drain register GIOPDR[7:0] register. If open-drain mode output is enabled on a pin, a high value (1) written to the data output register (GIDOUTx[7:0]) forces the pin to a high impedance state (Z).

- Pull disable (GIOPULDIS)
Disables the internal pull on GIO terminal(s) configured as inputs by writing to the GIOPULDISx registers.
- Pull select (GIOPSL)
Selects internal pull down (default) or pull up on GIO terminal(s) configured as inputs by writing to the GIOPULSELx registers.

Refer to the specific device's datasheet to identify the number of GIO ports as well as the input and output functions supported. Some devices may not support the programmable pull controls. In that case, the pull disable and the pull select register controls will not work.

25.3.2 Interrupt Function

The GIO module supports up to 32 terminals to be configured for generating an interrupt to the host processor through the Vectored Interrupt Manager (VIM). The main functions of the interrupt block are:

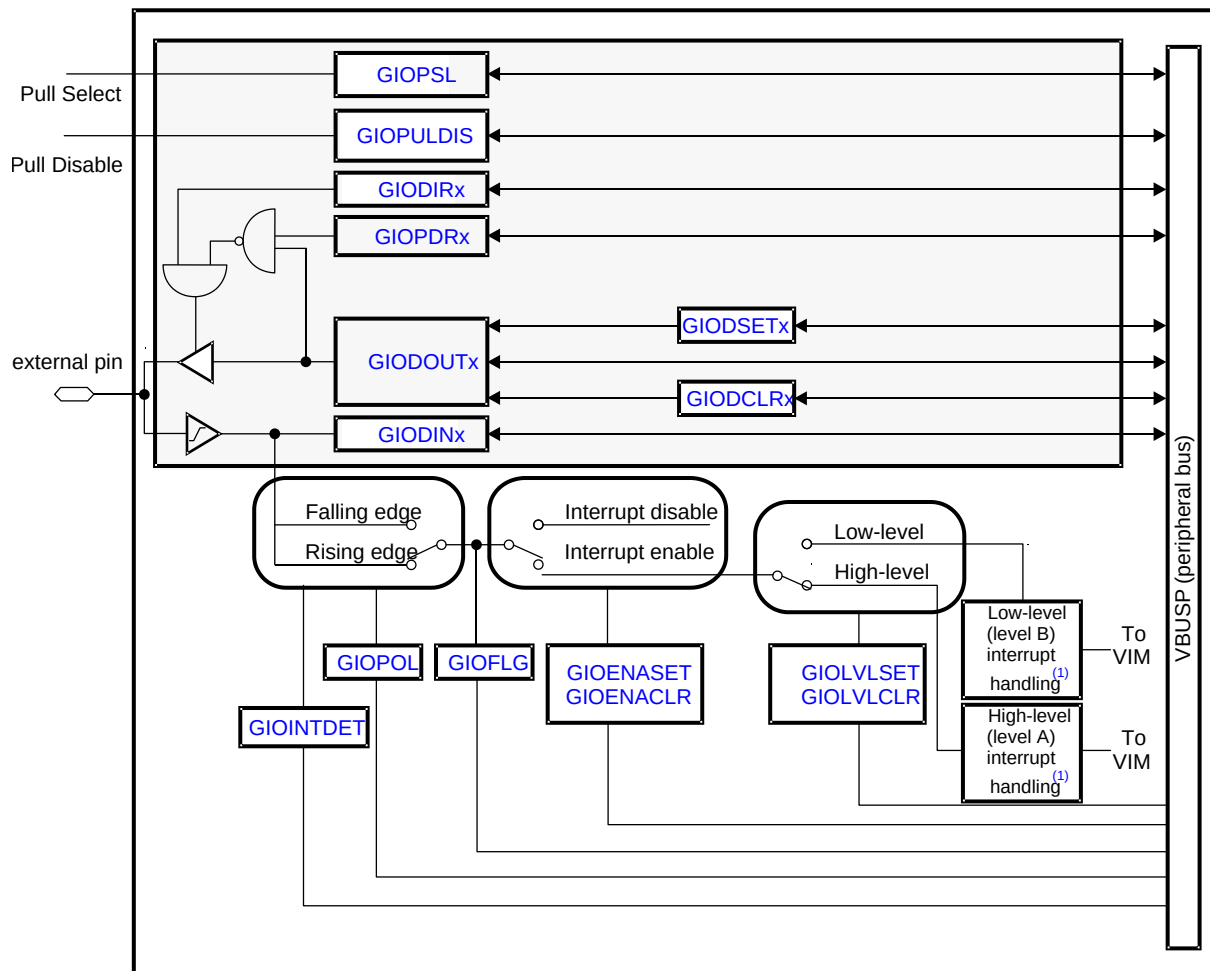
- Select the GIO pin(s) that is/are used to generate interrupt(s)
This is done via the interrupt enable set and clear registers, GIOENASET and GIOENACLR.
- Select the edge on the selected GIO pin(s) that is/are used to generate interrupt(s): rising/falling/both
Rising or falling edge can be selected via the GIOPOL register. If interrupt is required to be generated on both rising and falling edges, this can be configured via the GIOINTDET register.
- Select the interrupt priority
Low- or high-level interrupt can be selected through the GIOLVLSET and GIOLVLCLR registers.
- Individual interrupt flags are set in the GIOFLG register

The terminals on GIO ports A through D are all interrupt-capable and can be used to handle either general I/O functions or interrupt requests. Each interrupt request can be connected to the VIM at one of two different levels – High (or A) and Low (or B), depending on the VIM channel number. The VIM has an inherent priority scheme so that a request on a lower number channel has a higher priority than a request on a higher number channel. Refer the device datasheet to identify the VIM channel numbers for the GIO level A and level B interrupt requests. Also note that the interrupt priority of level A and level B interrupt handling blocks can be re-programmed in the VIM.

25.3.3 GIO Block Diagram

The GIO block diagram ([Figure 25-4](#)) represents the flow of information through a pin. The shaded area corresponds to the I/O block; the unshaded area corresponds to the interrupt block.

Figure 25-4. GIO Block Diagram



- (1) A single low-level-interrupt-handling block and a single high-level-interrupt-handling block service all of the interrupt-capable external pins, but only one pin can be serviced by an interrupt block at a time.

25.4 Device Modes of Operation

The GIO module behaves differently in different modes of operation. There are two main modes:

- Emulation mode
- Power-down mode (low-power mode)

25.4.1 Emulation Mode

Emulation mode is used by debugger tools to stop the CPU at breakpoints to read registers.

NOTE: Emulation Mode and Emulation Registers

Emulation mode is a mode of operation of the device and is separate from the GIO emulation registers (GIOEMU1 and GIOEMU2). The contents of these emulation registers are identical to the contents of GIO offset registers (GIOOFF1 and GIOOFF2). Both emulation registers and GIO offset registers are NOT cleared when they are read in emulation mode. GIO offset registers are cleared when they are read in normal mode (other than emulation mode). The emulation registers are NOT cleared when they are read in normal mode. The intention for the emulation registers is that software can use them without clearing the flags.

During emulation mode:

- External interrupts are not captured because the VIM is unable to service interrupts.
- Any register can be read without affecting the state of the system.
- A write to a register still does affect the state of the system.

25.4.2 Power-Down Mode (Low-Power Mode)

In power-down mode, the clock signal to the GIO module is disabled. Thus, there is no switching and the only current draw comes from leakage current. In power-down mode, interrupt pins become level-sensitive rather than edge-sensitive. The polarity bit changes function from falling-edge-triggered to low-level-triggered and rising-edge-triggered to high-level-triggered. A corresponding level on an interrupt pin pulls the module out of low-power mode, if the interrupt is also enabled to wake up the device out of a low-power mode.

25.4.2.1 Module-Level Power Down

The GIO module can be placed into a power down state by disabling the GIO peripheral module via the appropriate bit in the peripheral power down register. Please refer to the Peripheral Central Resource Registers ([Section 2.5.3](#)) for details.

25.4.2.2 Device-Level Power Down

The entire device can be placed in one of the pre-defined low-power modes: doze, snooze, or sleep using the clock source and clock domain disable registers in the system module.

25.5 GIO Control Registers

[Table 25-1](#) shows the summary of the GIO registers. The registers are accessible in 8-, 16-, and 32-bit reads or writes.

The start address for the GIO module is FFF7 BC00h.

The GIO module supports up to 8 ports. Refer to your device-specific data manual to identify the actual number of GIO ports and the number of pins in each GIO port implemented on this device.

The GIO module supports up to 4 interrupt-capable ports. Refer to the device datasheet to identify the actual number of interrupt-capable GIO ports and the number of pins in each GIO port implemented on this device.

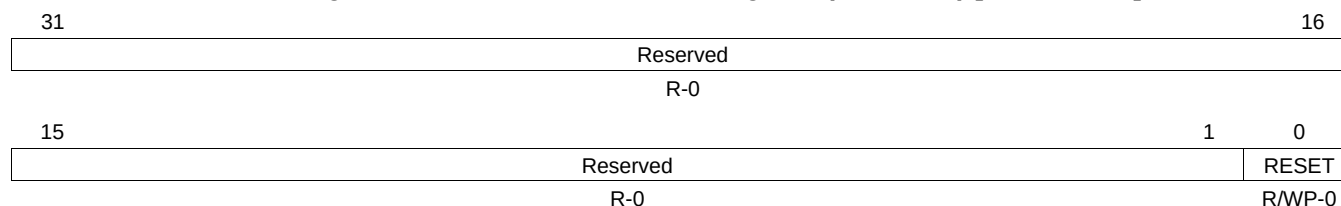
Table 25-1. GIO Control Registers

Offset	Acronym	Register Description	Section
00h	GIOGCR0	GIO Global Control Register	Section 25.5.1
08h	GIOINTDET	GIO Interrupt Detect Register	Section 25.5.2
0Ch	GIOPOL	GIO Interrupt Polarity Register	Section 25.5.3
10h	GIOENASET	GIO Interrupt Enable Set Register	Section 25.5.4.1
14h	GIOENACLR	GIO Interrupt Enable Clear Register	Section 25.5.4.2
18h	GIOLVLSET	GIO Interrupt Priority Set Register	Section 25.5.5.1
1Ch	GIOLVLCLR	GIO Interrupt Priority Clear Register	Section 25.5.5.2
20h	GIOFLG	GIO Interrupt Flag Register	Section 25.5.6
24h	GIOOFF1	GIO Offset 1 Register	Section 25.5.7
28h	GIOOFF2	GIO Offset 2 Register	Section 25.5.8
2Ch	GIOEMU1	GIO Emulation 1 Register	Section 25.5.9
30h	GIOEMU2	GIO Emulation 2 Register	Section 25.5.10
34h	GIODIRA	GIO Data Direction Register	Section 25.5.11
38h	GIODINA	GIO Data Input Register	Section 25.5.12
3Ch	GIODOUTA	GIO Data Output Register	Section 25.5.13
40h	GIODSETA	GIO Data Set Register	Section 25.5.14
44h	GIODCLRA	GIO Data Clear Register	Section 25.5.15
48h	GIOPDRA	GIO Open Drain Register	Section 25.5.16
4Ch	GIOPULDISA	GIO Pull Disable Register	Section 25.5.17
50h	GIOPSLA	GIO Pull Select Register	Section 25.5.18
54h	GIODIRB	GIO Data Direction Register	Section 25.5.11
58h	GIODINB	GIO Data Input Register	Section 25.5.12
5Ch	GIODOUTB	GIO Data Output Register	Section 25.5.13
60h	GIODSETB	GIO Data Set Register	Section 25.5.14
64h	GIODCLRB	GIO Data Clear Register	Section 25.5.15
68h	GIOPDRB	GIO Open Drain Register	Section 25.5.16
6Ch	GIOPULDISB	GIO Pull Disable Register	Section 25.5.17
70h	GIOPSLB	GIO Pull Select Register	Section 25.5.18

25.5.1 GIO Global Control Register (GIOGCR0)

The GIOGCR0 register contains one bit that controls the module reset status. Writing a 0 to this bit puts the module in a reset state. After system reset, this bit must be set to 1 before configuring any other register of the GIO module. [Figure 25-5](#) and [Table 25-2](#) describe this register.

Figure 25-5. GIO Global Control Register (GIOGCR0) [offset = 00h]



LEGEND: R/W = Read/Write; R = Read only; WP = Write in privileged mode only; -n = value after reset

Table 25-2. GIO Global Control Register (GIOGCR0) Field Descriptions

Bit	Field	Value	Description
31-1	Reserved	0	Reads return 0. Writes have no effect.
0	RESET	0	GIO reset.
		0	The GIO is in reset state.
		1	The GIO is operating normally.

NOTE: Note that putting the GIO module in reset state is not the same as putting it in a low-power state.

25.5.2 GIO Interrupt Detect Register (GIOINTDET)

The GIO module supports generation of an interrupt request to CPU when a rising edge, falling edge, or both edges is detected on one or more GIO pin(s). The GIOINTDET register allows both rising and falling edges to be detected, while the GIOPOL register allows the application to define whether a rising edge or a falling edge is to be detected. [Figure 25-6](#) and [Table 25-3](#) describe this register.

Figure 25-6. GIO Interrupt Detect Register (GIOINTDET) [offset = 08h]

31	24	23	16
GIOINTDET 3		GIOINTDET 2	
R/W-0		R/W-0	
15	8	7	0
GIOINTDET 1		GIOINTDET 0	
R/W-0		R/W-0	

LEGEND: R/W = Read/Write; -n = value after reset

Table 25-3. GIO Interrupt Detect Register (GIOINTDET) Field Descriptions

Bit	Field	Value	Description
31-24	GIOINTDET 3		Interrupt detection select for pins GIOD[7:0]
		0	The flag sets on either a falling or a rising edge on the corresponding pin, depending on the polarity setup in the polarity register (GIOPOL).
		1	The flag sets on both the rising and falling edges on the corresponding pin.
23-16	GIOINTDET 2		Interrupt detection select for pins GIOC[7:0]
		0	The flag sets on either a falling or a rising edge on the corresponding pin, depending on the polarity setup in the polarity register (GIOPOL).
		1	The flag sets on both the rising and falling edges on the corresponding pin.
15-8	GIOINTDET 1		Interrupt detection select for pins GIOB[7:0]
		0	The flag sets on either a falling or a rising edge on the corresponding pin, depending on the polarity setup in the polarity register (GIOPOL).
		1	The flag sets on both the rising and falling edges on the corresponding pin.
7-0	GIOINTDET 0		Interrupt detection select for pins GIOA[7:0]
		0	The flag sets on either a falling or a rising edge on the corresponding pin, depending on the polarity setup in the polarity register (GIOPOL).
		1	The flag sets on both the rising and falling edges on the corresponding pin.

25.5.3 GIO Interrupt Polarity Register (GIOPOL)

The GIOPOL register configures the polarity of the edge, rising edge or falling edge, that needs to be detected. When the device is in low-power mode, the GIOPOL register controls the **level**, high or low, which will be detected by the GIO module. [Figure 25-7](#) and [Table 25-4](#) describe this register.

Figure 25-7. GIO Interrupt Polarity Register (GIOPOL) [offset = 0Ch]

31	24	23	16
GIOPOL 3		GIOPOL 2	
R/W-0		R/W-0	
15	8	7	0
GIOPOL 1		GIOPOL 0	
R/W-0		R/W-0	

LEGEND: R/W = Read/Write; -n = value after reset

Table 25-4. GIO Interrupt Polarity Register (GIOPOL) Field Descriptions

Bit	Field	Value	Description
31-24	GIOPOL 3		Interrupt polarity select for pins GIOD[7:0] Normal operation (user or privileged mode):
		0	The flag is set on the falling edge on the corresponding pin.
		1	The flag is set on the rising edge on the corresponding pin.
			Low-power mode (GIO module clocks off):
23-16	GIOPOL 2	0	The interrupt is triggered on the low level.
		1	The interrupt is triggered on the high level.
			Low-power mode (GIO module clocks off):
		0	The interrupt is triggered on the low level.
15-8	GIOPOL 1	1	The interrupt is triggered on the high level.
			Interrupt polarity select for pins GIOB[7:0] Normal operation (user or privileged mode):
		0	The flag is set on the falling edge on the corresponding pin.
		1	The flag is set on the rising edge on the corresponding pin.
7-0	GIOPOL 0		Low-power mode (GIO module clocks off):
		0	The interrupt is triggered on the low level.
		1	The interrupt is triggered on the high level.
			Interrupt polarity select for pins GIOA[7:0] Normal operation (user or privileged mode):
		0	The flag is set on the falling edge on the corresponding pin.
		1	The flag is set on the rising edge on the corresponding pin.
			Low-power mode (GIO module clocks off):
		0	The interrupt is triggered on the low level.
		1	The interrupt is triggered on the high level.

25.5.4 GIO Interrupt Enable Registers (GIOENASET and GIOENACLR)

The GIOENASET and GIOENACLR registers control which interrupt-capable pins are actually configured as interrupts. If the interrupt is enabled, the rising edge, falling edge, or both edges on the selected pin lead to an interrupt request.

25.5.4.1 GIOENASET Register

Figure 25-8 and Table 25-5 describe this register.

NOTE: Enabling Interrupt at the Device Level

The interrupt channel in the Vectored Interrupt Manager (VIM) must be enabled for the interrupt request to be forwarded to the CPU. Additionally, the ARM CPU (CPSR bit 7 or 6) must be cleared to respond to interrupt requests (IRQ/FIQ).

Figure 25-8. GIO Interrupt Enable Set Register (GIOENASET) [offset = 10h]

31	24	23	16
GIOENASET 3		GIOENASET 2	
R/W-0		R/W-0	
15	8	7	0
GIOENASET 1		GIOENASET 0	
R/W-0		R/W-0	

LEGEND: R/W = Read/Write; -n = value after reset

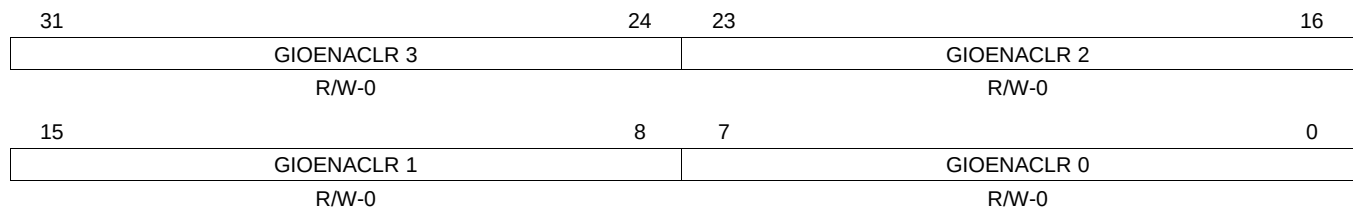
Table 25-5. GIO Interrupt Enable Set Register (GIOENASET) Field Descriptions

Bit	Field	Value	Description
31-24	GIOENASET 3	0	Interrupt enable for pins GIOD[7:0] Read: The interrupt is disabled. Write: Writing a 0 to this bit has no effect.
		1	Read: The interrupt is enabled. Write: Enables the interrupt.
23-16	GIOENASET 2	0	Interrupt enable for pins GIOC[7:0] Read: The interrupt is disabled. Write: Writing a 0 to this bit has no effect.
		1	Read: The interrupt is enabled. Write: Enables the interrupt.
15-8	GIOENASET 1	0	Interrupt enable for pins GIOB[7:0] Read: The interrupt is disabled. Write: Writing a 0 to this bit has no effect.
		1	Read: The interrupt is enabled. Write: Enables the interrupt.
7-0	GIOENASET 0	0	Interrupt enable for pins GIOA[7:0] Read: The interrupt is disabled. Write: Writing a 0 to this bit has no effect.
		1	Read: The interrupt is enabled. Write: Enables the interrupt.

25.5.4.2 GIOENACLR Register

This register disables the interrupt. [Figure 25-9](#) and [Table 25-6](#) describe this register.

Figure 25-9. GIO Interrupt Enable Clear Register (GIOENACLR) [offset = 14h]



LEGEND: R/W = Read/Write; -n = value after reset

Table 25-6. GIO Interrupt Enable Clear Register (GIOENACLR) Field Descriptions

Bit	Field	Value	Description
31-24	GIOENACLR 3	0	Interrupt disable for pins GIOD[7:0] Read: The interrupt is disabled. Write: Writing a 0 to this bit has no effect.
		1	Read: The interrupt is enabled. Write: Disables the interrupt.
23-16	GIOENACLR 2	0	Interrupt disable for pins GIOC[7:0] Read: The interrupt is disabled. Write: Writing a 0 to this bit has no effect.
		1	Read: The interrupt is enabled. Write: Disables the interrupt.
15-8	GIOENACLR 1	0	Interrupt disable for pins GIOB[7:0] Read: The interrupt is disabled. Write: Writing a 0 to this bit has no effect.
		1	Read: The interrupt is enabled. Write: Disables the interrupt.
7-0	GIOENACLR 0	0	Interrupt disable for pins GIOA[7:0] Read: The interrupt is disabled. Write: Writing a 0 to this bit has no effect.
		1	Read: The interrupt is enabled. Write: Disables the interrupt.

25.5.5 GIO Interrupt Priority Registers (GIOLVLSET and GIOLVLCLR)

The GIOLVLSET and GIOLVLCLR registers configure the interrupts as high-level (level A) or low-level (level B) going to the Vectored Interrupt Manager (VIM). Each interrupt is individually configured.

- The high-level interrupts are recorded to GIOFF1 and GIOEMU1.
- The low-level interrupts are recorded to GIOFF2 and GIOEMU2.

NOTE: The GIO module can generate two interrupt requests. These are connected to two separate channels on the Vectored Interrupt Manager (VIM). The lower-numbered VIM channels are higher priority. The GIO interrupt connected to a lower-number channel is the high-level (also called level A) GIO interrupt, while the GIO interrupt connected to a higher-number channel is the low-level (also called level B) GIO interrupt.

25.5.5.1 GIOLVLSET Register

The GIOLVLSET register is used to configure an interrupt as a high-level interrupt going to the VIM. An interrupt can be configured as a high-level interrupt by writing a 1 into the corresponding bit of the GIOLVLSET register. Writing a 0 has no effect. [Figure 25-10](#) and [Table 25-7](#) describe this register.

Figure 25-10. GIO Interrupt Priority Register (GIOLVLSET) [offset = 18h]

31											16												
GIOLVLSET 3											GIOLVLSET 2												
R/W-0											R/W-0												
15											8	7											0
GIOLVLSET 1											GIOLVLSET 0												
R/W-0											R/W-0												

LEGEND: R/W = Read/Write; -n = value after reset

Table 25-7. GIO Interrupt Priority Register (GIOLVLSET) Field Descriptions

Bit	Field	Value	Description
31-24	GIOLVLSET 3	0	GIO high-priority interrupt for pins GIOD[7:0]. Read: The interrupt is a low-level interrupt. The low-level interrupts are recorded to GIOFF2 and GIOEMU2. Write: Writing a 0 to this bit has no effect.
		1	Read: The interrupt is set as a high-level interrupt. The high-level interrupts are recorded to GIOFF1 and GIOEMU1. Write: Sets the interrupt as a high-level interrupt. The high-level interrupts are recorded to GIOFF1 and GIOEMU1.
23-16	GIOLVLSET 2	0	GIO high-priority interrupt for pins GIOC[7:0]. Read: The interrupt is a low-level interrupt. The low-level interrupts are recorded to GIOFF2 and GIOEMU2. Write: Writing a 0 to this bit has no effect.
		1	Read: The interrupt is set as a high-level interrupt. The high-level interrupts are recorded to GIOFF1 and GIOEMU1. Write: Sets the interrupt as a high-level interrupt. The high-level interrupts are recorded to GIOFF1 and GIOEMU1.
15-8	GIOLVLSET 1	0	GIO high-priority interrupt for pins GIOB[7:0]. Read: The interrupt is a low-level interrupt. The low-level interrupts are recorded to GIOFF2 and GIOEMU2. Write: Writing a 0 to this bit has no effect.
		1	Read: The interrupt is set as a high-level interrupt. The high-level interrupts are recorded to GIOFF1 and GIOEMU1. Write: Sets the interrupt as a high-level interrupt. The high-level interrupts are recorded to GIOFF1 and GIOEMU1.

Table 25-7. GIO Interrupt Priority Register (GIOLVLSET) Field Descriptions (continued)

Bit	Field	Value	Description
7-0	GIOLVLSET 0	0	GIO high-priority interrupt for pins GIOA[7:0]. Read: The interrupt is a low-level interrupt. The low-level interrupts are recorded to GIOOFF2 and GIOEMU2. Write: Writing a 0 to this bit has no effect.
		1	Read: The interrupt is set as a high-level interrupt. The high-level interrupts are recorded to GIOOFF1 and GIOEMU1. Write: Sets the interrupt as a high-level interrupt. The high-level interrupts are recorded to GIOOFF1 and GIOEMU1.

25.5.5.2 GIOLVLCLR Register

The GIOLVLCLR register is used to configure an interrupt as a low-level interrupt going to the VIM. An interrupt can be configured as a low-level interrupt by writing a 1 into the corresponding bit of the GIOLVLCLR register. Writing a 0 has no effect. [Figure 25-11](#) and [Table 25-8](#) describe this register.

Figure 25-11. GIO Interrupt Priority Register (GIOLVLCLR) [offset = 1Ch]

31	GIOLVLCLR 3								16
	R/W-0								
	GIOLVLCLR 2								
	R/W-0								
15	GIOLVLCLR 1				8	7	GIOLVLCLR 0		
	R/W-0						R/W-0		

LEGEND: R/W = Read/Write; -n = value after reset

Table 25-8. GIO Interrupt Priority Register (GIOLVLCLR) Field Descriptions

Bit	Field	Value	Description
31-24	GIOLVLCLR 3	0	GIO low-priority interrupt for pins GIOD[7:0] Read: The interrupt is a low-level interrupt. Write: Writing a 0 to this bit has no effect.
		1	Read: The interrupt is set as a high-level interrupt. The high-level interrupts are recorded to GIOFF1 and GIOEMU1. Write: Sets the interrupt as a low-level interrupt. The low-level interrupts are recorded to GIOFF2 and GIOEMU2.
23-16	GIOLVLCLR 2	0	GIO low-priority interrupt for pins GIOC[7:0] Read: The interrupt is a low-level interrupt. Write: Writing a 0 to this bit has no effect.
		1	Read: The interrupt is set as a high-level interrupt. The high-level interrupts are recorded to GIOFF1 and GIOEMU1. Write: Sets the interrupt as a low-level interrupt. The low-level interrupts are recorded to GIOFF2 and GIOEMU2.
15-8	GIOLVLCLR 1	0	GIO low-priority interrupt for pins GIOB[7:0] Read: The interrupt is a low-level interrupt. Write: Writing a 0 to this bit has no effect.
		1	Read: The interrupt is set as a high-level interrupt. The high-level interrupts are recorded to GIOFF1 and GIOEMU1. Write: Sets the interrupt as a low-level interrupt. The low-level interrupts are recorded to GIOFF2 and GIOEMU2.
7-0	GIOLVLCLR 0	0	GIO low-priority interrupt for pins GIOA[7:0] Read: The interrupt is a low-level interrupt. Write: Writing a 0 to this bit has no effect.
		1	Read: The interrupt is set as a high-level interrupt. The high-level interrupts are recorded to GIOFF1 and GIOEMU1. Write: Sets the interrupt as a low-level interrupt. The low-level interrupts are recorded to GIOFF2 and GIOEMU2.

25.5.6 GIO Interrupt Flag Register (GIOFLG)

The GIOFLG register contains flags indicating that the transition edge (as set in GIOINTDET and GIOPOL registers) has occurred. The flag can be cleared by the CPU writing a 1 to the flag that is set. The flag is also cleared by reading the appropriate interrupt offset register (GIOOFF1 or GIOOFF2). [Figure 25-12](#) and [Table 25-9](#) describe this register.

Figure 25-12. GIO Interrupt Flag Register (GIOFLG) [offset = 20h]

31	24	23	16
GIOFLG 3		GIOFLG 2	
R/W1C-0		R/W1C-0	
15	8	7	0
GIOFLG 1		GIOFLG 0	
R/W1C-0		R/W1C-0	

LEGEND: R/W = Read/Write; W1C = Write 1 to clear; -n = value after reset

Table 25-9. GIO Interrupt Flag Register (GIOFLG) Field Descriptions

Bit	Field	Value	Description
31-24	GIOFLG 3	0	GIO flag for pins GIOD[7:0] Read: A transition has not occurred since the last clear. Write: Writing a 0 to this bit has no effect.
		1	Read: The selected transition on the corresponding pin has occurred. Write: The corresponding bit is cleared to 0. Note: This bit is also cleared by a read to the corresponding bit in the appropriate offset register.
23-16	GIOFLG 2	0	GIO flag for pins GIOC[7:0] Read: A transition has not occurred since the last clear. Write: Writing a 0 to this bit has no effect.
		1	Read: The selected transition on the corresponding pin has occurred. Write: The corresponding bit is cleared to 0. Note: This bit is also cleared by a read to the corresponding bit in the appropriate offset register.
15-8	GIOFLG 1	0	GIO flag for pins GIOB[7:0] Read: A transition has not occurred since the last clear. Write: Writing a 0 to this bit has no effect.
		1	Read: The selected transition on the corresponding pin has occurred. Write: The corresponding bit is cleared to 0. Note: This bit is also cleared by a read to the corresponding bit in the appropriate offset register.
7-0	GIOFLG 0	0	GIO flag for pins GIOA[7:0] Read: A transition has not occurred since the last clear. Write: Writing a 0 to this bit has no effect.
		1	Read: The selected transition on the corresponding pin has occurred. Write: The corresponding bit is cleared to 0. Note: This bit is also cleared by a read to the corresponding bit in the appropriate offset register.

NOTE: An interrupt flag gets set when the selected transition happens on the corresponding GIO pin **regardless of whether the interrupt generation is enabled or not**. It is recommended to clear a flag before enabling the interrupt generation for a transition on the corresponding GIO pin.

25.5.7 GIO Offset Register 1 (GIOOFF1)

The GIOFF1 register provides a numerical offset value that represents the pending external interrupt with high priority. The offset value can be used to locate the position of the interrupt routine in a vector table in application software. [Figure 25-13](#) and [Table 25-10](#) describe this register.

NOTE: Reading this register clears it, GIOEMU1 and the corresponding flag bit in the GIOFLG register. However, in emulation mode, a read to this register does not clear any register or flag. If more than one GIO interrupts are pending, then reading the GIOOFF1 register will change the contents of GIOOFF1 and GIOEMU1 to show the offset value for the next highest-priority pending interrupt. The application can choose to service all GIO interrupts from the same service routine by continuing to read the GIOOFF1 register until it reads zeros.

Figure 25-13. GIO Offset 1 Register (GIOOFF1) [offset = 24h]

31			16
Reserved			
R-0			
15	6	5	0
Reserved		GIOFF1	
R-0		R-0	

LEGEND: R = Read only; -n = value after reset

Table 25-10. GIO Offset 1 Register (GIOOFF1) Field Descriptions

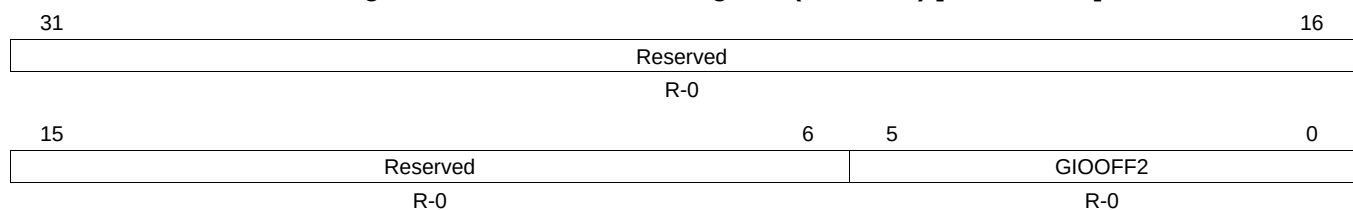
Bit	Field	Value	Description
31-6	Reserved	0	Reads return 0. Writes have no effect.
5-0	GIOFF1		GIO offset 1. These bits index the currently pending high-priority interrupt. This register and the flag bit (in the GIOFLG register) are also cleared when this register is read, except in emulation mode.
		0	No interrupt is pending.
		1h	Interrupt 0 (corresponding to GIOA0) is pending with a high priority.
		:	:
		8h	Interrupt 7 (corresponding to GIOA7) is pending with a high priority.
		9h	Interrupt 8 (corresponding to GIOB0) is pending with a high priority.
		:	:
		10h	Interrupt 16 (corresponding to GIOB7) is pending with a high priority.
		:	:
		20h	Interrupt 32 (corresponding to GIOD7) is pending with a high priority.
		21h-3Fh	Reserved

25.5.8 GIO Offset B Register (GIOFF2)

The GIOFF2 register provides a numerical offset value that represents the pending external interrupt with low priority. The offset value can be used to locate the position of the interrupt routine in a vector table in application software. [Figure 25-14](#) and [Table 25-11](#) describe this register.

NOTE: Reading this register clears it, GIOEMU2 and the corresponding flag bit in the GIOFLG register. However, in emulation mode, a read to this register does not clear any register or flag. If more than one GIO interrupts are pending, then reading the GIOFF1 register will change the contents of GIOFF2 and GIOEMU2 to show the offset value for the next highest-priority pending interrupt. The application can choose to service all GIO interrupts from the same service routine by continuing to read the GIOFF1 register until it reads zeros.

Figure 25-14. GIO Offset 2 Register (GIOFF2) [offset = 28h]



LEGEND: R = Read only; -n = value after reset

Table 25-11. GIO Offset 2 Register (GIOFF2) Field Descriptions

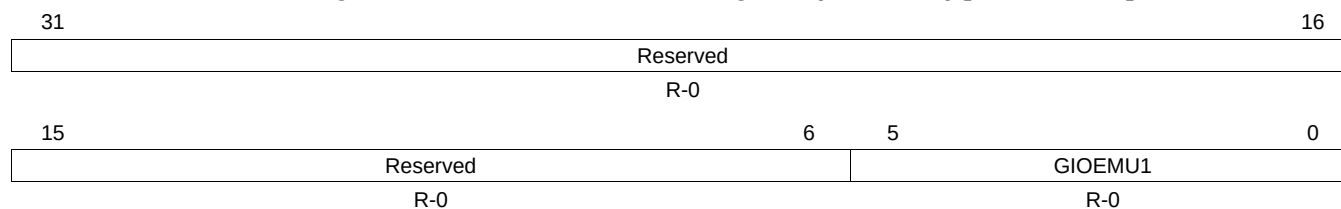
Bit	Field	Value	Description
31-6	Reserved	0	Reads return 0. Writes have no effect.
5-0	GIOFF2		GIO offset 2. These bits index the currently pending low-priority interrupt. This register and the flag bit (in the GIOFLG register) are also cleared when this register is read, except in emulation mode.
		0	No interrupt is pending.
		1h	Interrupt 0 (corresponding to GIOA0) is pending with a low priority.
		:	:
		8h	Interrupt 7 (corresponding to GIOA7) is pending with a low priority.
		9h	Interrupt 8 (corresponding to GIOB0) is pending with a low priority.
		:	:
		10h	Interrupt 16 (corresponding to GIOB7) is pending with a low priority.
		:	:
		20h	Interrupt 32 (corresponding to GIOD7) is pending with a low priority.
		21h-3Fh	Reserved

25.5.9 GIO Emulation A Register (GIOEMU1)

The GIOEMU1 register is a read-only register. The contents of this register are identical to the contents of GIOOFF1. The intention for the this register is that software can use it without clearing the flags. [Figure 25-15](#) and [Table 25-12](#) describe this register.

NOTE: The corresponding flag in the GIOFLG register is not cleared when the GIOEMU1 register is read.

Figure 25-15. GIO Emulation 1 Register (GIOEMU1) [offset = 2Ch]



LEGEND: R = Read only; -n = value after reset

Table 25-12. GIO Emulation 1 Register (GIOEMU1) Field Descriptions

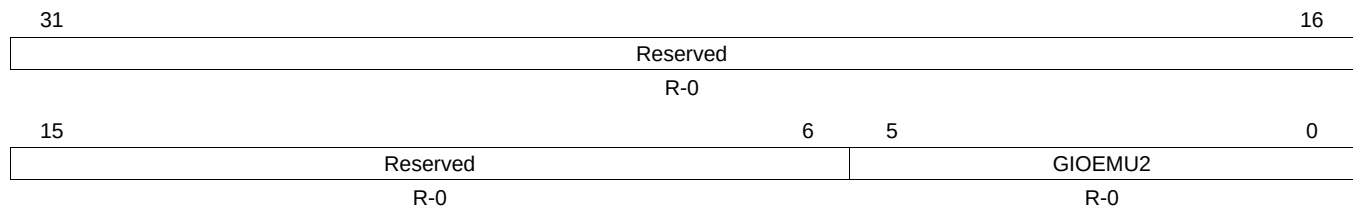
Bit	Field	Value	Description
31-6	Reserved	0	Reads return 0. Writes have no effect.
5-0	GIOEMU1		GIO offset emulation 1. These bits index the currently pending high-priority interrupt. No register or flag is cleared by reading this register.
		0	No interrupt is pending.
		1h	Interrupt 0 (corresponding to GIOA0) is pending with a high priority.
		:	:
		8h	Interrupt 7 (corresponding to GIOA7) is pending with a high priority.
		9h	Interrupt 8 (corresponding to GIOB0) is pending with a high priority.
		:	:
		10h	Interrupt 16 (corresponding to GIOB7) is pending with a high priority.
		:	:
		20h	Interrupt 32 (corresponding to GIOD7) is pending with a high priority.
		21h-3Fh	Reserved

25.5.10 GIO Emulation B Register (GIOEMU2)

The GIOEMU2 register is a read-only register. The contents of this register are identical to the contents of GIOOFF2. The intention for this register is that software can use it without clearing the flags. [Figure 25-16](#) and [Table 25-13](#) describe this register.

NOTE: The corresponding flag in the GIOFLG register is not cleared when the GIOEMU2 register is read.

Figure 25-16. GIO Emulation 2 Register (GIOEMU2) [offset = 30h]



LEGEND: R = Read only; -n = value after reset

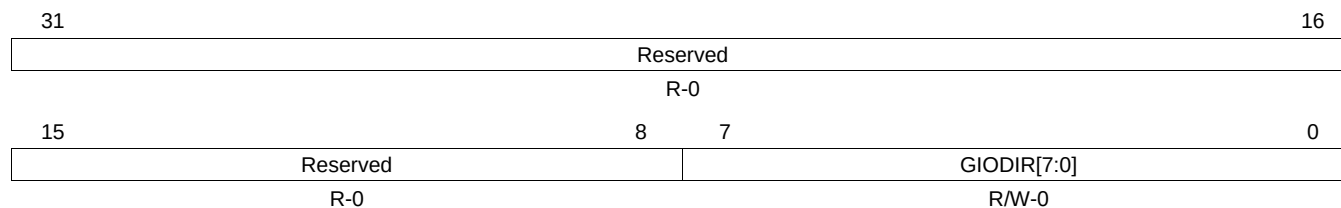
Table 25-13. GIO Emulation 2 Register (GIOEMU2) Field Descriptions

Bit	Field	Value	Description
31-6	Reserved	0	Reads return 0. Writes have no effect.
5-0	GIOEMU2	0	GIO offset emulation 2. These bits index the currently pending low-priority interrupt. No register or flag is cleared by reading this register.
		1h	No interrupt is pending.
		:	Interrupt 0 (corresponding to GIOA0) is pending with a low priority.
		:	:
		8h	Interrupt 7 (corresponding to GIOA7) is pending with a low priority.
		9h	Interrupt 8 (corresponding to GIOB0) is pending with a low priority.
		:	:
		10h	Interrupt 16 (corresponding to GIOB7) is pending with a low priority.
		:	:
		20h	Interrupt 32 (corresponding to GIOD7) is pending with a low priority.
		21h-3Fh	Reserved

25.5.11 GIO Data Direction Registers (GIODIR[A-B])

The GIODIR register controls whether the pins of a given port are configured as inputs or outputs. [Figure 25-17](#) and [Table 25-14](#) describe this register.

Figure 25-17. GIO Data Direction Registers (GIODIR[A-B]) [offset = 34h, 54h]



LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

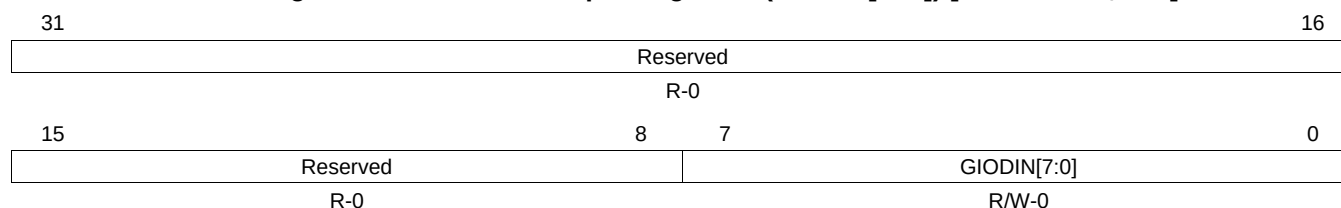
Table 25-14. GIO Data Direction Registers (GIODIR[A-B]) Field Descriptions

Bit	Field	Value	Description
31-8	Reserved	0	Reads return 0. Writes have no effect.
7-0	GIODIR[n]	0	GIO data direction of port n, pins [7:0] The GIO pin is an input. Note: If the pin direction is set as an input, the output buffer is tristated.
		1	The GIO pin is an output.

25.5.12 GIO Data Input Registers (GIODIN[A-B])

Values in the GIODIN register reflect the current state (high = 1 or low = 0) on the pins of the port. [Figure 25-18](#) and [Table 25-15](#) describe this register.

Figure 25-18. GIO Data Input Registers (GIODIN[A-B]) [offset = 38h, 58h]



LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 25-15. GIO Data Input Registers (GIODIN[A-B]) Field Descriptions

Bit	Field	Value	Description
31-8	Reserved	0	Reads return 0. Writes have no effect.
7-0	GIODIN[n]	0	GIO data input for port n, pins [7:0] The pin is at logic low (0).
		1	The pin is at logic high (1).

25.5.13 GIO Data Output Registers (GIODOUT[A-B])

Values in the GIODOUT register specify the output state (high = 1 or low = 0) of the pins of the port when they are configured as outputs. [Figure 25-19](#) and [Table 25-16](#) describe this register.

NOTE: Values in the GIODSET register set the data output control register bits to 1 regardless of the current value in the GIODOUT bits.

Figure 25-19. GIO Data Output Registers (GIODOUT[A-B]) [offset = 3Ch, 5Ch]

31	Reserved																16
R-0																	
15	Reserved							8	7	GIODOUT[7:0]							0
R-0									R/W-0								

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 25-16. GIO Data Output Registers (GIODOUT[A-B]) Field Descriptions

Bit	Field	Value	Description
31-8	Reserved	0	Reads return 0. Writes have no effect.
7-0	GIODOUT[n]	0	GIO data output of port n, pins[7:0]. The pin is driven to logic low (0).
		1	The pin is driven to logic high (1). Note: Output is in high impedance state if the GIOPDRx bit = 1 and GIODOUTx bit = 1. Note: GIO pin is placed in output mode by setting the GIODIRx bit to 1.

25.5.14 GIO Data Set Registers (GIODSET[A-B])

Values in this register set the data output control register bits to 1 regardless of the current value in the GIODOUT bits. The contents of this register reflect the contents of GIODOUT. [Figure 25-20](#) and [Table 25-17](#) describe this register.

Figure 25-20. GIO Data Set Registers (GIODSET[A-B]) [offset = 40h, 60h]

31																	16
Reserved																	
R-0																	
15								8	7								0
Reserved								GIODSET[7:0]									
R-0								R/W-0									

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 25-17. GIO Data Set Registers (GIODSET[A-B]) Field Descriptions

Bit	Field	Value	Description
31-8	Reserved	0	Reads return 0. Writes have no effect.
7-0	GIODSET[n]	0	GIO data set for port n, pins[7:0]. This bit drives the output of GIO pin high. Write: Writing a 0 has no effect.
		1	Write: The corresponding GIO pin is driven to logic high (1). Note: The current logic state of the GIODOUT bit will also be displayed by this bit. Note: GIO pin is placed in output mode by setting the GIODIRx bit to 1.

25.5.15 GIO Data Clear Registers (GIODCLR[A-B])

Values in this register clear the data output register (GIO Data Output Register [A-H]) bit to 0 regardless of its current value. The contents of this register reflect the contents of GIODOUT. [Figure 25-21](#) and [Table 25-18](#) describe this register.

Figure 25-21. GIO Data Clear Registers (GIODCLR[A-B]) [offset = 44h, 64h]

31																16	
Reserved																	
R-0																	
15								8	7								0
Reserved								GIODCLR[7:0]									
R-0								R/W-0									

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 25-18. GIO Data Clear Registers (GIODCLR[A-B]) Field Descriptions

Bit	Field	Value	Description
31-8	Reserved	0	Reads return 0. Writes have no effect.
7-0	GIODCLR[n]	0	GIO data clear for port n, pins[7:0]. This bit drives the output of GIO pin low. Write: Writing a 0 has no effect.
		1	Write: The corresponding GIO pin is driven to logic low (0). Note: The current logic state of the GIODOUT bit will also be displayed by this bit. Note: GIO pin is placed in output mode by setting the GIODIRx bit to 1.

25.5.16 GIO Open Drain Registers (GIOPDR[A-B])

Values in this register enable or disable the open drain capability of the data pins. [Figure 25-22](#) and [Table 25-19](#) describe this register.

Figure 25-22. GIO Open Drain Registers (GIOPDR[A-B]) [offset = 48h, 68h]

31																16	
Reserved																	
R-0																	
15								8	7								0
Reserved								GIOPDR[7:0]									
R-0								R/W-0									

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 25-19. GIO Open Drain Registers (GIOPDR[A-B]) Field Descriptions

Bit	Field	Value	Description
31-8	Reserved	0	Reads return 0. Writes have no effect.
7-0	GIOPDR[n]	0	GIO open drain for port n, pins[7:0] The GIO pin is configured in push/pull (normal GIO) mode. The output voltage is V _{OL} or lower if GIODOUT bit = 0 and V _{OH} or higher if GIODOUT bit = 1.
		1	The GIO pin is configured in open drain mode. The GIODOUTx bit controls the state of the GIO output buffer: GIODOUTx = 0, the GIO output buffer is driven low; GIODOUTx = 1, the GIO output buffer is tristated.

25.5.17 GIO Pull Disable Registers (GIOPULDIS[A-B])

Values in this register enable or disable the pull control capability of the pins. [Figure 25-23](#) and [Table 25-20](#) describe this register.

Figure 25-23. GIO Pull Disable Registers (GIOPULDIS[A-B]) [offset = 4Ch, 6Ch]

31											16
Reserved											
R-0											
15					8	7					0
Reserved						GIOPULDIS[7:0]					
R-0						R/W-0					

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 25-20. GIO Pull Disable Registers (GIOPULDIS[A-B]) Field Descriptions

Bit	Field	Value	Description
31-8	Reserved	0	Reads return 0. Writes have no effect.
7-0	GIOPULDIS[n]	0	GIO pull disable for port n, pins[7:0]. Writes to this bit will only take effect when the GIO pin configured as an input pin.
		1	The pull functionality is disabled.
			Note: The GIO pin is placed in input mode by clearing the GIODIRx bit to 0.

25.5.18 GIO Pull Select Registers (GIOPSL[A-B])

Values in this register select the pull up or pull down functionality of the pins. [Figure 25-24](#) and [Table 25-21](#) describe this register.

Figure 25-24. GIO Pull Select Registers (GIOPSL[A-B]) [offset = 50h, 70h]

31											16
Reserved											
R-0											
15					8	7					0
Reserved					GIOPSL[7:0]						
R-0					R/W-0						

LEGEND: R/W = Read/Write; R = Read only; -n = value after reset

Table 25-21. GIO Pull Select Registers (GIOPSL[A-B]) Field Descriptions

Bit	Field	Value	Description
31-8	Reserved	0	Reads return 0. Writes have no effect.
7-0	GIOPSL[n]	0	GIO pull select for port n, pins[7:0]
		1	The pull down functionality is select, when pull up/pull down logic is enabled.
		1	The pull up functionality is select, when pull up/pull down logic is enabled.
			Note: The pull up/pull down functionality is enabled by clearing corresponding bit in GIOPULDIS to 0.

25.6 I/O Control Summary

The behavior of the output buffer and the pull control is summarized in [Table 25-22](#).

Table 25-22. Output Buffer and Pull Control Behavior for GIO Pins

Module under Reset?	Pin Direction (GDIR) ⁽¹⁾⁽²⁾	Open Drain Enable (GODR) ⁽¹⁾⁽³⁾	Pull Disable (GODIS) ⁽¹⁾⁽⁴⁾	Pull Select (GOSL) ⁽¹⁾⁽⁵⁾	Pull Control	Output Buffer ⁽⁶⁾
Yes	X	X	X	X	Enabled	Disabled
No	0	X	0	0	Pull down	Disabled
No	0	X	0	1	Pull up	Disabled
No	0	X	1	0	Disabled	Disabled
No	0	X	1	1	Disabled	Disabled
No	1	0	X	X	Disabled	Enabled
No	1	1	X	X	Disabled	Enabled

⁽¹⁾ X = Don't care

⁽²⁾ GDIR = 0 for input; = 1 for output

⁽³⁾ See [Section 25.5.16](#)

⁽⁴⁾ GODIS = 0 for enabling pull control; = 1 for disabling pull control

⁽⁵⁾ GOSL = 0 for pull-down functionality; = 1 for pull-up functionality

⁽⁶⁾ If open drain is enabled, output buffer will be disabled if a high level (1) is being output.